Salvage Drop - Technical Sheet

Tech used

One self-contained HTML file (about 39 KB): HTML5 Canvas 2D rendering, vanilla JavaScript, no libraries, no external assets.

Web Audio API: all sound effects and the adaptive soundtrack are synthesized live (oscillators plus a filtered noise buffer). Audio is optional and unlocks on the first tap, so browsers without audio still run.

Touch Events with dual floating virtual joysticks and large 76 px action buttons; mouse and keyboard fallbacks. Safe-area insets for notched phones; device pixel ratio capped at 2 for performance.

Fixed 60 Hz simulation step separate from rendering; localStorage save.

Game systems

Seeded procedural map (2400x2400): landing zone, 3 nests, 2 data cores, 10 caches and 70 rocks placed with spacing rules.

Squad AI: bots orbit the player, gather on active cores, kite melee enemies, revive the downed player and return as reinforcements.

Threat director: threat = win tier + time + nests destroyed + extraction pressure. It drives spawn rate, wave size, enemy HP scaling and enemy type mix (crawler, spitter, brute). Enemy cap of 60.

Abilities: Shock Pulse (area damage, knockback, invulnerability) and Orbital Strike (delayed area blast with friendly fire).

Extraction loop: objectives, then beacon, 60 s hold, 20 s boarding window. Salvage economy with 6 upgrade tracks (cost grows x1.6 per level).

Graphics: neon glow shading, particle bursts, shockwave rings, screenshake, impact flashes, damage numbers, animated landing pad and dropship, off-screen objective arrows, minimap, low-HP vignette.

Music: tempo rises with threat and the key drops during extraction.

Audit 1 - Balance (simulated play)

A headless autopilot played full missions using the exact game code. First version: 0 wins in 20 runs - the squad wiped in about 90 s, and losing paid nothing, so there was no progression.

Fixes: enemy damage x0.35, enemy HP x0.7, slower spawns (9 s base), slower threat growth, 2 revives instead of 1, and losses now keep 25% of salvage.

Result: fresh-save win rate 50% (avg mission about 3 min); in a 10-mission campaign buying upgrades, win rate went from 80% in the first half to 100% in the second. A win-tier threat increase was added so later missions keep getting harder.

Audit 2 - Bugs

Smoke test ran the full app (menu, deploy, 400 s of play with abilities, armory, menu) against a stubbed browser with no errors.

Fixed: a salvage pickup exactly on the player could divide by zero and produce an invalid position; pickups now collect before the magnet pull.

Fixed: dying with one revive left ended the mission immediately because the revive counter check was too strict (raised to 2 revives).

Audio context is resumed on tap for iOS, and the game auto-pauses when the tab is hidden.

Audit 3 - Player experience

Objective panel always says the next step; off-screen arrows and a minimap point to nests, cores and the LZ.

The Call Ship button only appears when it can be used, so there is no dead button.

Toasts announce nests, cores, downed squadmates and the dropship; a red pulse warns at low HP; the strike shows its blast ring before it lands.

Losing still pays 25%, so every run moves you forward.

Testing note

Tested by simulated play and a stubbed-browser smoke test, not on a physical phone.